Republic of the Philippines
CAVITE STATE UNIVERSITY
Don Severino de las Alas Campus
Indang, Cavite

COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY
Department of Information Technology

Name: _____ Student No.: _____
Course / Yr. / Sec.: _____ Date: _____

Portfolio #2 – Gaps of Security and Technology

Objectives

- To identify the security gaps created when business priorities prioritize speed and openness over technical controls.
- To map specific vulnerabilities to their corresponding modern threats and malware types.
- To evaluate the impact of security gaps on the four primary functions of Information Security within an organization.

Background / Scenario

This is an individual assessment.

You are an Information Security Consultant hired to audit "NextGen Logistics," a fast-growing delivery company. The company recently upgraded its technology to improve delivery speeds, but they failed to update their security practices, creating massive "gaps" between their technology and their security.

During your audit, you discovered the following critical gaps:

- Gap 1: To allow remote work, the company lets employees download the company's financial database onto their personal, unmonitored home laptops so they can work offline.
- Gap 2: The company installed hundreds of new IoT smart-thermostats in their warehouses to save energy. However, the IT team left the default manufacturer passwords (admin/admin) on all devices to make them "easier to manage."
- Gap 3: The HR department frequently posts high-resolution photos of new employees holding their company ID badges on the public company Facebook page to show off their "open culture."

Instructions

Part 1: The Security Trinity & Malware Analysis

Analyze **Gap 1** and **Gap 2** using the definitions discussed in the lecture (Threat, Vulnerability, Attack).

- a. For **Gap 1**, identify the specific *Vulnerability*. If a hacker uses this vulnerability to install a program that encrypts the financial database and demands Bitcoin, what specific type of malware is this?
- b. For **Gap 2**, identify the specific *Vulnerability*. If a hacker takes control of these thermostats to launch a volumetric attack against a competitor's website, what specific type of modern attack is this (e.g., APT, DDoS)? Explain how the thermostats are used in this attack.

Part 2: Social Engineering Analysis

Analyze **Gap 3**.

- a. Explain how a malicious hacker could use the information in these Facebook photos to conduct a **Pretexting** attack against the company's IT Helpdesk.
- b. Explain how an attacker could use the same photos to execute a **Tailgating (Piggybacking)** attack at the physical warehouse.

Part 3: Business Impact Evaluation
Choose *one* of the gaps above and explain how an attack exploiting it would directly violate at least **two** of the "Four Primary Functions of InfoSec" (e.g., Protects ability to function, Enables safe operations, Protects data, Safeguards tech assets).

Reflection

1. Reviewing the "Conflict" between Business and Security discussed in the lecture, why do you think companies like NextGen Logistics frequently leave "gaps" in their security when adopting new, convenient technologies?

Submission Format

SUBMIT YOUR ANSWERS in a single PDF document and remove all headers and rubric in your submission. Filename: **Portfolio1_Surname_Section.pdf** (e.g., Portfolio1_DelaCruz_BSCS-3-1.pdf). Include all members' names and their individual reflections at the end of the document.

Rubric

Criterion	Excellent (21-25 Points)	Good (16-20 Points)	Needs Improvement (11-15 Points)	Poor (0-10 Points)	Points
1. Following Instructions & Formatting	All parts of the activity are fully completed. The PDF is correctly named and perfectly structured with all required sections clearly labeled.	All major parts are completed, but there is a minor omission or a slight deviation from the required format.	A significant portion of the submission is incomplete, or the document is poorly formatted making it difficult to read.	The activity is not submitted, or it completely disregards the instructions and format.	/25
2. AI Originality & Effort	The submission demonstrates clear, original thought. Explanations are in the student's own words and apply the lecture concepts accurately. No signs of AI generation.	The work appears to be the student's own, but the analysis is somewhat generic. The language is simple and lacks boilerplate phrasing.	The submission is vague, overly formal, or contains generic phrasing that raises concerns about originality and the potential use of AI tools for generation.	The submission is clearly generated by an AI tool with little to no modification, is plagiarized, or shows a severe lack of effort.	/25
3. Threat & Malware Mapping (Part 1 & 2)	Flawlessly identifies the vulnerabilities. Accurately identifies the malware (Ransomware), attack type (DDoS/Botnet), and Social Engineering tactics (Pretexting/Tailgating) with precise explanations.	Correctly identifies the vulnerabilities and attack types, but the explanations of their mechanisms lack some technical depth.	Confuses threats with vulnerabilities. Misidentifies the malware type or provides an incorrect definition of how the social engineering attack operates.	Fails to identify the core concepts. Explanations are entirely incorrect or missing.	/25
4. Business Impact & Reflection	Deeply understands the Four Primary Functions of InfoSec and accurately applies them to the chosen gap. Reflection provides profound insight into the business vs. security conflict.	Identifies the correct InfoSec functions but explanations are brief. Reflection is relevant and shows a good understanding of theory.	Struggles to map the attack to the specific functions of InfoSec. Reflection attempts to answer the prompt but demonstrates confusion.	Analysis of the business impact is fundamentally flawed. Reflection questions are left blank or are irrelevant to the topic.	/25
TOTAL SCORE					/100